

Tempus AI

NASDAQ : TEM · \$13.06B mkt cap · 180M sh · \$644M cash, \$560M debt · \$1.3B revenue FY25 (+83%, Ambry-boosted); FY26 guide \$1.59-1.60B (+25%);
Mature- Adj-EBITDA inflected positive (FY26 guide ~\$65M) but SBC ~\$136M/yr → owner-FCF still negative; Net cash ~\$84M (\$0.644B cash vs
Company DCF \$0.560B Ares term loan); Lefkofsky ~60% voting control

\$72.69

THE CENTRAL QUESTION

At ~6.6x sales and near-breakeven, is the market paying for proven cash economics, or for a data-moat plus a margin inflection that SBC currently swamps and the modal case doesn't deliver?

Tempus AI is AI-precision-oncology diagnostics plus a multimodal health-data flywheel. Revenue is \$1.3B, growing 25-36%, adj-EBITDA just inflected positive (FY26 guide ~\$65M). But SBC runs ~\$136M/yr — more than 10% of revenue and larger than that adj-EBITDA — so true owner free cash flow is still negative. In July 2026 Tempus agreed to buy the rest of Personalis for ~\$1.5B, mostly stock — entering MRD via NeXT Personal, but a ~15% share issuance dilutes the modal case. At \$46.93 the modal DCF is \$26.43, ~44% below spot; only the bull case clears the price. You are paying for hypergrowth and an unproven margin turn.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$29.72

expected fair value today
-59.1% vs spot \$72.69

FORWARD COMPOUNDED VALUE

+5y	+10y	+15y	+20y
\$47.12	\$74.74	\$118.62	\$188.39
0.65x spot	1.03x spot	1.63x spot	2.59x spot

WEIGHTING RATIONALE

Ultra Bear 14% Growth fades to ~8%, margin tops out ~5%, distressed re-rate; DCF \$3.00.

Bear 28% ~12% growth, pricing pressure, margins ~11% but thin; DCF \$11.73.

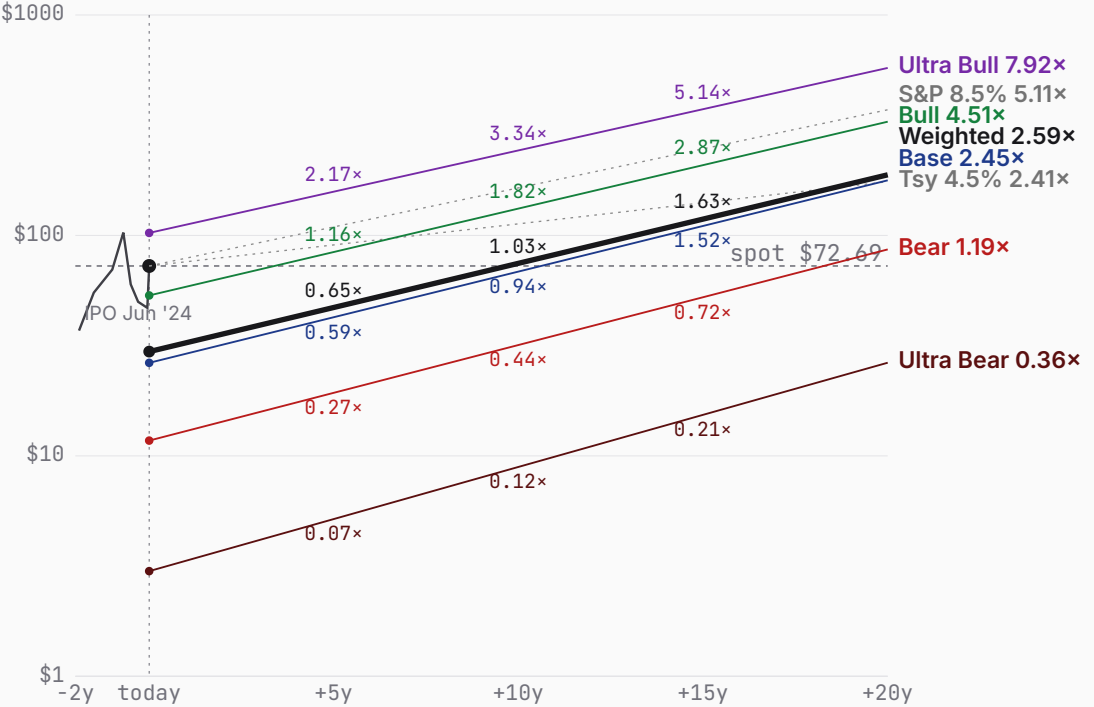
Base 33% ~16% growth, data line scales, margins ~17%; DCF \$26.43 — below spot.

Bull 17% ~21% growth plus a ~23% terminal margin; DCF \$53.43 — clears spot.

Ultra Bull 8% ~25% growth, software margins, data standard; DCF \$102.66.

Bull (17%) + Ultra Bull (8%) together contribute \$17.30 — 58% of the \$29.72 expected value despite 25% combined probability. Today's spot (\$72.69) sits 145% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$3.00	BEAR	\$11.73	BASE	\$26.43	BULL	\$53.43	ULTRA BULL	\$102.66
	-95.9% vs spot		-83.9% vs spot		-63.6% vs spot		-26.5% vs spot		+41.2% vs spot
CAGR +8.1% WACC 11.5% CAGR +8.1% Prob 14%		CAGR +12.1% WACC 10.5% CAGR +12.1% Prob 28%		CAGR +16.3% WACC 10.0% CAGR +16.3% Prob 33%		CAGR +20.6% WACC 9.5% CAGR +20.6% Prob 17%		CAGR +25.2% WACC 9.0% CAGR +25.2% Prob 8%	
Growth collapses, never durably profitable		Decelerates hard, thin margins persist		Solid growth, but modal value sits below spot		Data flywheel drives a real margin inflection		Becomes the oncology-data standard	

AR
2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

Tempus AI scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 14% Bear 28% Base 33% Bull 17% Ultra Bull 8% · Weighted expected \$29.72 (-59.1% vs spot)

ULTRA BEAR	\$3.00	BEAR	\$11.73	BASE	\$26.43	BULL	\$53.43	ULTRA BULL	\$102.66
	-95.9% vs spot		-83.9% vs spot		-63.6% vs spot		-26.5% vs spot		+41.2% vs spot

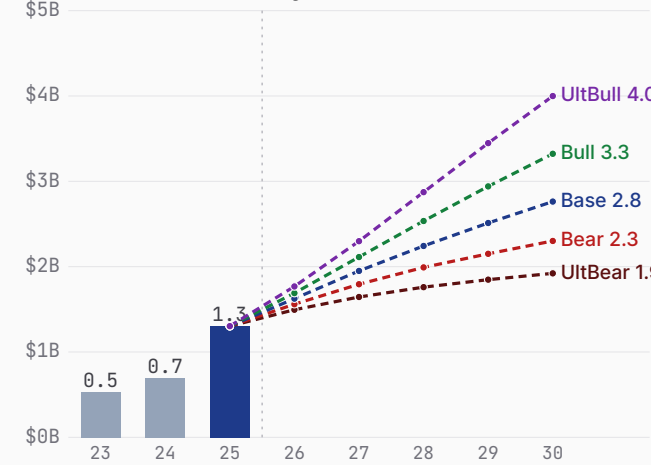
Probability 14%	Probability 28%	Probability 33%	Probability 17%	Probability 8%
Growth collapses, never durably profitable WHY 14% A ~14% tail. Requires both the diagnostics franchise to stall and Data and Applications to stop scaling — possible if pharma budgets tighten and foundation-model rivals commoditize the data, but it contradicts the +40%+ data-licensing trend now in evidence. WHAT HAPPENS Revenue growth fades toward ~8% as oncology test volume saturates and the data-licensing pipeline stalls. The margin path inches from ~8% to only 5% — operating leverage never arrives because SBC and clinical-lab costs scale with revenue. The company de-rates to a distressed multiple as the market stops paying for a moat that did not compound. Dilution from continued stock comp erodes per-share value. The DCF lands at \$3.00, roughly 94% below spot.	Decelerates hard, thin margins persist WHY 28% A ~28% weight, the second-likeliest outcome. This is the world where TEM is a good but ordinary diagnostics-plus-data business — real growth, real competition, no durable software margin. Pricing pressure is the swing variable. WHAT HAPPENS Growth slows to ~12% as Guardant, Natera, and Caris pressure diagnostics pricing and reimbursement caps revenue per test; Natera's Signatera leads the MRD race Tempus just bought into. The data flywheel keeps compounding but cannot offset a maturing core diagnostics line. Margins climb from ~4% to ~11% — positive, but far short of software economics, because SBC and lab infrastructure stay heavy and the Personalis integration adds near-term losses. The DCF is \$11.73, about 75% below spot. The market re-rates toward a mid-teens-growth diagnostics comp.	Solid growth, but modal value sits below spot WHY 33% The single most-likely path at ~33%. It assumes the current trajectory broadly holds — mid-teens growth, a scaling data line, gradual margin gains — without the step-change in operating leverage the bull requires. It is the honest anchor for the finding. WHAT HAPPENS Revenue compounds ~16% as oncology volume grows ~28% and Data and Applications scales the ~80%-margin pharma-licensing line. The AstraZeneca-Pathos deal type repeats, and the 7.3M-record library keeps feeding both segments. Margins ramp from ~2% to ~17% as the data mix lifts the blend and the May-2026 FDA tumor-only xT CDx approval adds an estimated ~\$200-per-test ASP uplift from 2027, but SBC keeps owner-FCF modest early. The DCF is \$26.43 — a healthy business, about 44% below spot once the ~15% Personalis share issuance is absorbed. The modal case simply does not support \$46.93.	Data flywheel drives a real margin inflection WHY 17% A ~17% weight. Requires two things at once — sustained high growth AND a genuine terminal margin, not just adj-EBITDA. Plausible if the data moat re-rates the whole company, but it is the entry hurdle, not the expected case. WHAT HAPPENS Growth holds at ~21% as Data and Applications compounds 40%+ and the ~80%-margin data line becomes a larger share of the mix. The multimodal library proves hard to replicate, NeXT Personal scales Tempus into MRD, and pharma standardizes on Tempus for oncology data. Operating leverage finally clears SBC: margins ramp from 2% to ~23%, aided by the xT CDx ASP uplift, reaching a durable-profit inflection. The DCF is \$53.43, about 14% above spot even after the Personalis dilution. This is the lowest scenario that justifies today's price.	Becomes the oncology-data standard WHY 8% An ~8% tail. The data-moat optionality the price already embeds, paid off in full: a network-and-scale flywheel that no rival replicates. Low probability, but it is the asymmetric reason a long position is debated at all. WHAT HAPPENS TEM wins the platform: the multimodal library becomes the default substrate for oncology drug development and clinical decisioning. Revenue compounds ~25%, with Data and Applications the growth engine at software gross margins. Margins ramp from 5% to ~29% as the high-margin data mix dominates and scale economics compound. The DCF reaches \$102.66, about 119% above spot — a platform-winner outcome, not a diagnostics multiple.

Tempus AI business snapshot

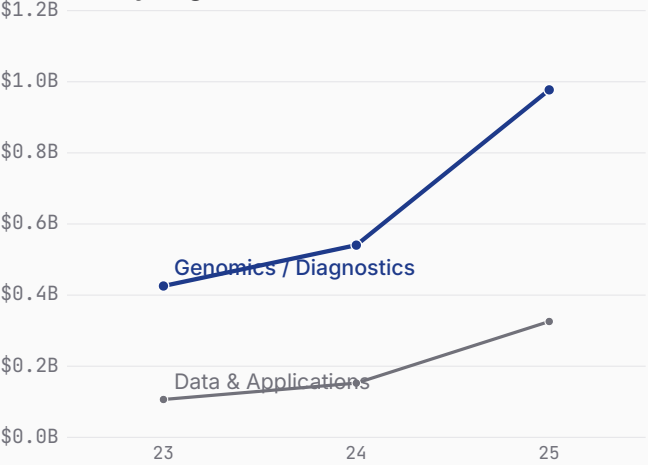
Bear \$11.73 Base \$26.43 Bull \$53.43

FY23-FY25 history + FY26-FY30 scenario projections · fiscal years end Dec 31 · FY2025 (Feb'26) + Q1'26 results

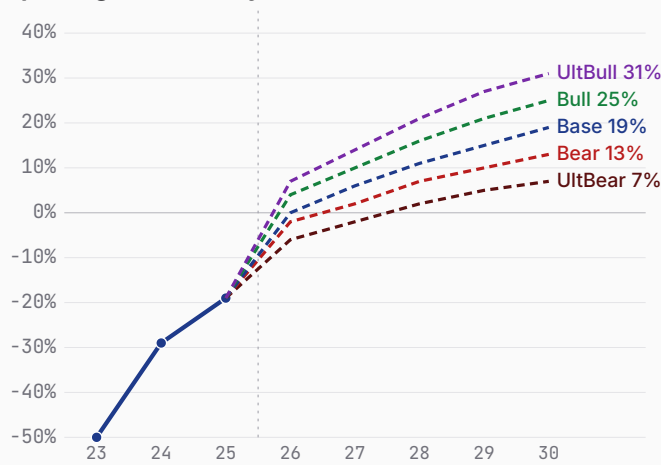
Revenue (\$B) — history + scenarios to FY30



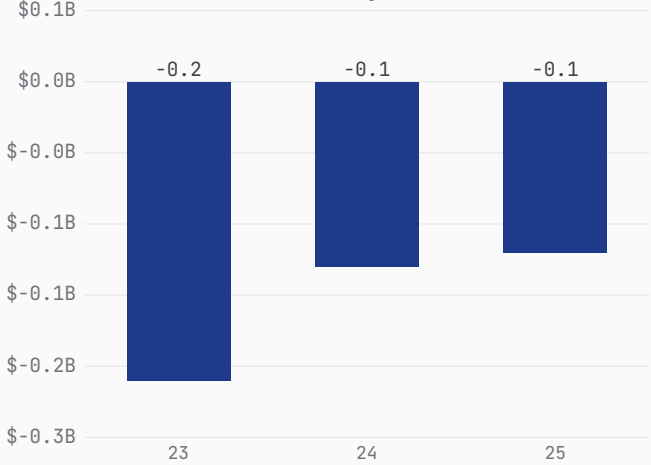
Revenue by segment (\$B)



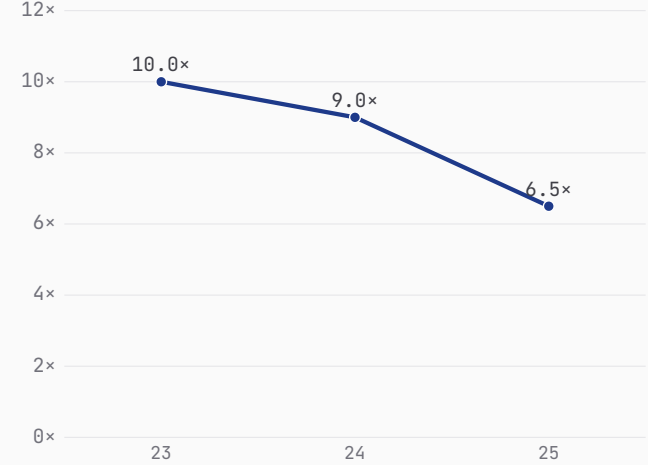
Op margin — history + scenarios



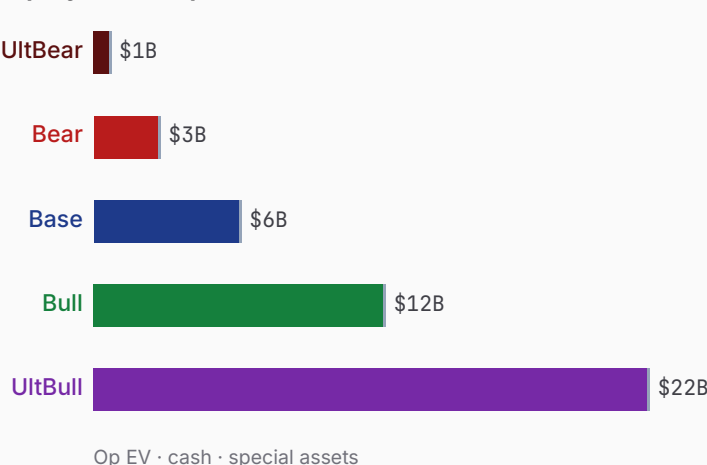
Free cash flow (\$B) — history



EV / Revenue — historical



Equity build (Op EV + net cash)



Sources: Tempus AI FY2025 10-K + Q1'26 8-K (CIK 1717115), FY2026 guidance. Revenue as a growth-rate path off FY25; FCF = revenue x an SBC-adjusted owner-FCF margin (the ~\$136M/yr SBC exceeds the +\$65M adj-EBITDA, so the modeled margin starts negative); Gordon terminal at scenario WACC. EV/sales vs precision-diagnostics peers.

Tempus AI 7 Powers · the moat, scored

Bear \$11.73 Base \$26.43 Bull \$53.43

Arena. AI-precision-oncology: molecular diagnostics, a multimodal-data-licensing flywheel, and — via the July-2026 Personalis acquisition and its NeXT Personal test — the fast-growing minimal-residual-disease (MRD) arena (a ~\$5-8B-by-2035 market), against Guardant, Natera, Caris, Foundation Medicine (Roche), and Exact Sciences.

POWER AUDIT
dominant-power durability: medium

POWER AUDIT · 7 POWERS

Scale Economies 	Larger record base lowers unit cost of model training and licensing; reinforces the data flywheel as volume grows.
Network Economies 	More clinical use feeds more records, which improve products that attract more use — a data network loop, not a user network.
Counter-Positioning 	Software-margin data licensing is a model legacy lab-only rivals can't easily adopt without cannibalizing test revenue.
Switching Costs 	Pharma data pipelines and clinician workflow integration create modest stickiness, but contracts are re-competed.
Branding 	Recognized in AI-oncology, but diagnostics buyers decide on accuracy, reimbursement, and price, not brand.
Cornered Resource · thesis leans here 	The 7.3M-record multimodal library (1.3M genomic, 1.4M imaging) is the proprietary, hard-to-replicate asset and the dominant Power; the Personalis acquisition folds NeXT Personal MRD and its clinical volume (~80% already routed through Tempus) into it.
Process Power 	Multimodal data engineering is a capability, but not yet a durable, uncopyable process advantage.

PEER SET

Guardant Health Liquid biopsy and MRD (Guardant Reveal); scaling oncology volume	20% gr · ~6x sales
Natera Direct MRD leader (Signatera PMA filed Feb 2026, raised 2026 outlook, Japan entry); near breakeven	30% gr · 0% mgn · ~9x sales
Foundation Medicine Roche-owned genomic-profiling division; deep-pocketed	· division of Roche
Exact Sciences Cologuard plus oncology MRD (Oncodetect); established but slower-growing	12% gr · ~4x sales

THREAT VECTORS · FALSIFIERS

cornered_resource erosion · foundation-model rivals
falsifier: Open or competitor-trained oncology models replicate the data edge, commoditizing the ~80%-margin licensing line.

scale_economies / pricing · Guardant, Natera, Caris
falsifier: Diagnostics price competition compresses revenue per test, starving the volume that feeds the data flywheel — sharper now that Tempus has bought directly into the MRD contest against Natera's Signatera.

regulatory / reimbursement · payers and CMS
falsifier: Reimbursement cuts to molecular diagnostics undermine the core economics and the funnel that grows the record library.

COMPETITIVE READ

A durable Power is plausibly present but not proven. The cornered resource — a 7.3M-record multimodal library feeding an ~80%-margin licensing flywheel with a data-network loop — is genuinely hard to replicate, hence the bull case. But durability is medium, not high: the falsifier is foundation models trained on open or rival data closing the gap, or diagnostics pricing pressure starving the volume that feeds it. The moat must keep Data and Applications compounding 40%+ to earn the price. The July-2026 Personalis acquisition extends the cornered resource into MRD (NeXT Personal) while buying directly into that price war.

Tempus AI show your work

Bear \$11.73 Base \$26.43 Bull \$53.43 · Weighted \$29.72 (-59.1%)

ULTRA BEAR\$3.00					BEAR\$11.73					BASE\$26.43					BULL\$53.43					ULTRA BULL\$102.66				
ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS					ASSUMPTIONS				
5y revenue CAGR (%)					+8.1					+12.1					+16.3					+20.6				
Year-5 op margin (%)					7.0					13.0					19.0					25.0				
WACC (%)					11.5					10.5					10.0					9.5				
Terminal growth (%)					3.0					3.5					4.0					4.5				
Terminal FCF (\$B)					\$0.10B					\$0.25B					\$0.47B					\$0.76B				
Starting revenue (\$B)					1.30					1.30					1.30					1.30				
Scenario probability (%)					14					28					33					17				
5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B					5-YEAR DCF · \$B				
FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF	FY	Rev	Margin	FCF	PV FCF
FY26	1.49	-6%	-0.12	-0.11	FY26	1.56	-2%	-0.06	-0.06	FY26	1.63	+0%	-0.03	-0.03	FY26	1.69	+4%	+0.03	+0.03	FY26	1.77	+7%	+0.09	+0.08
FY27	1.64	-2%	-0.07	-0.05	FY27	1.79	+2%	+0.00	+0.00	FY27	1.95	+6%	+0.08	+0.06	FY27	2.11	+10%	+0.17	+0.14	FY27	2.30	+14%	+0.28	+0.23
FY28	1.76	+2%	+0.00	+0.00	FY28	1.99	+7%	+0.10	+0.07	FY28	2.24	+11%	+0.20	+0.15	FY28	2.54	+16%	+0.35	+0.27	FY28	2.87	+21%	+0.55	+0.42
FY29	1.85	+5%	+0.06	+0.04	FY29	2.15	+10%	+0.17	+0.12	FY29	2.51	+15%	+0.33	+0.22	FY29	2.94	+21%	+0.56	+0.39	FY29	3.45	+27%	+0.86	+0.61
FY30	1.92	+7%	+0.10	+0.06	FY30	2.30	+13%	+0.25	+0.15	FY30	2.76	+19%	+0.47	+0.29	FY30	3.32	+25%	+0.76	+0.48	FY30	4.00	+31%	+1.16	+0.75
Σ PV explicit FCF					-0.07					Σ PV explicit FCF					Σ PV explicit FCF					Σ PV explicit FCF				
+ PV terminal (g 3.0%)					0.68					+ PV terminal (g 3.5%)					+ PV terminal (g 4.0%)					+ PV terminal (g 4.5%)				
EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE					EQUITY BUILD · \$B & PER SHARE				
Operating EV					\$0.61B					Operating EV					Operating EV					Operating EV				
+ Cash & investments					\$0.64B					+ Cash & investments					+ Cash & investments					+ Cash & investments				
– Total debt					\$0.56B					– Total debt					– Total debt					– Total debt				
= Equity value					\$0.69B					= Equity value					= Equity value					= Equity value				
FY30 shares (M)					230					FY30 shares (M)					FY30 shares (M)					FY30 shares (M)				
= Expected per share					\$3.00					= Expected per share					= Expected per share					= Expected per share				
FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT					FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT				
+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y		+5y	+10y	+15y	+20y	
\$5.17	\$8.91	\$15.35	\$26.46		\$19.32	\$31.84	\$52.45	\$86.41		\$42.57	\$68.55	\$110.40	\$177.81		\$84.11	\$132.41	\$208.45	\$328.15		\$157.96	\$243.03	\$373.94	\$575.35	
0.07×	0.12×	0.21×	0.36×		0.27×	0.44×	0.72×	1.19×		0.59×	0.94×	1.52×	2.45×		1.16×	1.82×	2.87×	4.51×		2.17×	3.34×	5.14×	7.92×	

Tempus AI

People · Opportunity · Context · Deal

Bear \$11.73 Base \$26.43 Bull \$53.43

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Eric Lefkofsky · founder-led

11 yrs in seat

24%insider ownership

HIGHkey-person risk

Capital allocation (realized)
Founder Eric Lefkofsky is Chairman & CEO (founded 2015; serial founder of Groupon, Lightbank, Mediaocean and Pathos AI). Capital funds rapid growth and acquisitions (Ambry Genetics ~\$672M, Paige, OneOme; and in July 2026 a definitive agreement to buy the ~88% of Personalis it does not own for ~\$1.5B, mostly stock — the largest deal to date and dilutive) against ~\$605M cash; FY2025 revenue \$1.27B (+83%) at roughly adjusted-EBITDA breakeven. No dividend or buyback, and the share count is rising from deal stock plus heavy stock-based comp — FY2025 SBC of ~\$125M far exceeds the ~\$7M adjusted-EBITDA loss, so owner cash economics are weaker than the adjusted figure implies.

Incentive alignment
Lefkofsky took no base salary in 2024 and an \$800K base + \$800K target from 2025; alignment is via his stake. But the related-party web is the standout flag: Tempus COO Ryan Fukushima simultaneously runs Lefkofsky-founded Pathos AI, which is contracted to pay Tempus ~\$200M over three years; there is also a SoftBank JV and possible affiliated landlord. Insider trading is programmatic 10b5-1 selling. An active securities suit alleges AI-washing / round-tripping (unadjudicated).

GOVERNANCE FLAGS

- multi-class super-voting: Class B = 30 votes per share; Lefkofsky controls ~59.9% of the vote on ~24% economics — majority single-person control
- combined Chair & CEO (Lefkofsky); no confirmed lead independent director
- the Class B sunset is person-tied to Lefkofsky (converts only when he exits or drops below 10M shares) — no fixed-date sunset
- dense related-party web around Lefkofsky entities (Pathos AI via a shared COO, ~\$200M contract; SoftBank JV) — the subject of an active securities suit
- stock-based comp (~\$125M FY2025) exceeds the adjusted-EBITDA result; no buyback to offset dilution
- mitigants: the controlled-company exemption was declined; 8 of 9 directors independent; board declassified (annual elections)

PEOPLE READ

The most-entrenched tier: founder-CEO Lefkofsky holds ~59.9% of the vote on ~24% economics through 30-vote Class B, combines Chair and CEO, and sits at the center of a dense related-party web (a shared COO running a Lefkofsky company contracted to pay Tempus ~\$200M) now under securities litigation. Real mitigants keep it a 2-with-mitigants rather than lower — the controlled-company exemption was declined, the board is 8-of-9 independent and declassified — but majority single-person control plus the related-party concentration drive the score. Key-person risk is high.

THE FOUR LEGS

People

observable composite · 1–5

Opportunity

The scenario distribution · the business snapshot · the Arthur Indicator

Context

The 7 Powers analysis (Power Audit)

Deal

Open-market common — entry price weighed against the probability-weighted fair value (the -37% finding) + position sizing.



Tempus AI supporting analysis · disclaimers · glossary

Bear \$11.73 Base \$26.43 Bull \$53.43

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

The moat is real and hard to replicate
7.3M multimodal records (1.3M genomic, 1.4M imaging) feeding an ~80%-margin licensing flywheel is a cornered resource rivals can't quickly assemble.
- 2

Data and Applications is compounding +40%+
If data-licensing keeps growing 44% as in Q1'26, the high-margin mix re-rates the whole company toward the bull/ultra-bull path.
- 3

Ambry adds scale and a funnel
The ~\$600M Ambry deal added hereditary-cancer testing volume that feeds more records into the library — scale economics, not just revenue.
- 4

Adj-EBITDA already inflected positive
Q4'25 was +\$12.9M and FY26 guides ~\$65M; the margin turn the base needs may already be underway, just masked by SBC.
- 5

Pharma is validating the data, not just buying tests
The \$200M AstraZeneca-Pathos foundation-model deal prices the data asset directly — a repeatable, capital-light revenue type.

FALSIFICATION TRIGGERS

Bull validation Data and Applications holds 40%+ growth; margins expand as SBC falls; new pharma deals sign; NeXT Personal MRD and the xT CDx ASP uplift scale from 2027.	Bear validation Revenue-per-test compresses under Guardant/Natera/Caris; growth slips to the low teens; SBC exceeds adj-EBITDA, owner-FCF stays negative; Personalis cedes MRD share to Signatera.	Reframe needed Foundation models on rival data erode the licensing premium, or reimbursement cuts collapse the moat; a mostly-cash Personalis deal or an over-priced one reframes the capital-allocation read.
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Forward-looking statements. Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.	Author may hold positions. The author may own, intend to acquire, or hold short exposure to \$TEM or related securities at any time. Positions may change without notice and without an update to this document. Do not assume the author's positions match the tone of the analysis.	Do your own research. Do not make investment decisions on the basis of this document. Consult a licensed financial professional and read the company's official SEC filings (10-K, 10-Q, 8-K, proxy) before forming any investment view. If you wouldn't trust your retirement to a chatbot, don't trust this either.

GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Multimodal data library — TEM's 7.3M+ de-identified patient records linking genomic, imaging, and clinical data — the proprietary asset feeding both diagnostics and licensing.	Data and Applications — The pharma data-licensing segment (\$87M in Q1'26, +41%); ~80% gross margin, the high-margin flywheel distinct from the diagnostics lab business.	MRD — Minimal residual disease — detecting trace cancer after treatment; a growth vector for oncology testing volume and a competitive battleground.
Adj-EBITDA vs owner-FCF — Adjusted EBITDA adds back ~\$136M/yr of stock comp; owner free cash flow does not, so positive adj-EBITDA still maps to negative cash today.	SBC — Stock-based compensation (~\$136M/yr, over 10% of revenue) — a real, recurring, dilutive cost that exceeds adj-EBITDA and swamps the margin story.	Dual-class control — Founder-CEO Eric Lefkowsky holds ~60% of the vote via super-voting shares, concentrating control — a governance flag.